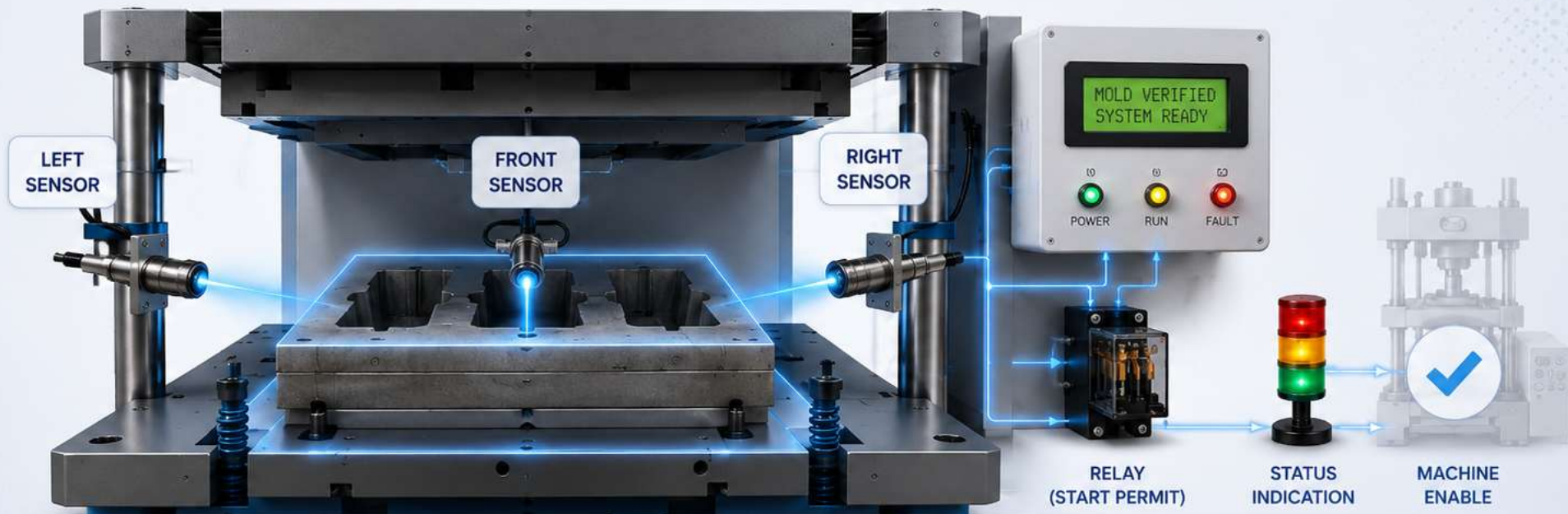


モールドけんしょう と あんぜんインターロックシステム Mold Verification and Safety Interlock System

Simulation-Based Industrial Safety Prototype



DEVELOPMENT ENVIRONMENT

- Proteus 8 Professional
- Arduino IDE



PROJECT FEATURES

- Three-Point Verification
- Safety Interlock
- Fault Detection
- Event Logging



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Industrial Challenge & Proposed Solution

! INDUSTRIAL CHALLENGES



Improper Mold Placement

Mold misalignment leads to defective products and equipment damage.



Unsafe Machine Start

Machines may start even when the mold is not correctly seated.



Manual Verification

Operator-dependent checks cause inconsistency and human error.



Production Downtime

Errors and rework result in delays and reduced productivity.



✓ OUR SOLUTION



Three-Point Verification

Three sensors ensure the mold is correctly seated at three critical points.



Safety Interlock

Machine starts only when all sensors confirm correct mold placement.



Fault Indication

Visual and audible alerts notify the operator during any fault condition.



Manual Reset Acknowledgement

System resumes operation only after the fault is cleared and reset is pressed.

どうきフロー Operational Workflow

Step-by-step operation of the mold verification and safety interlock system



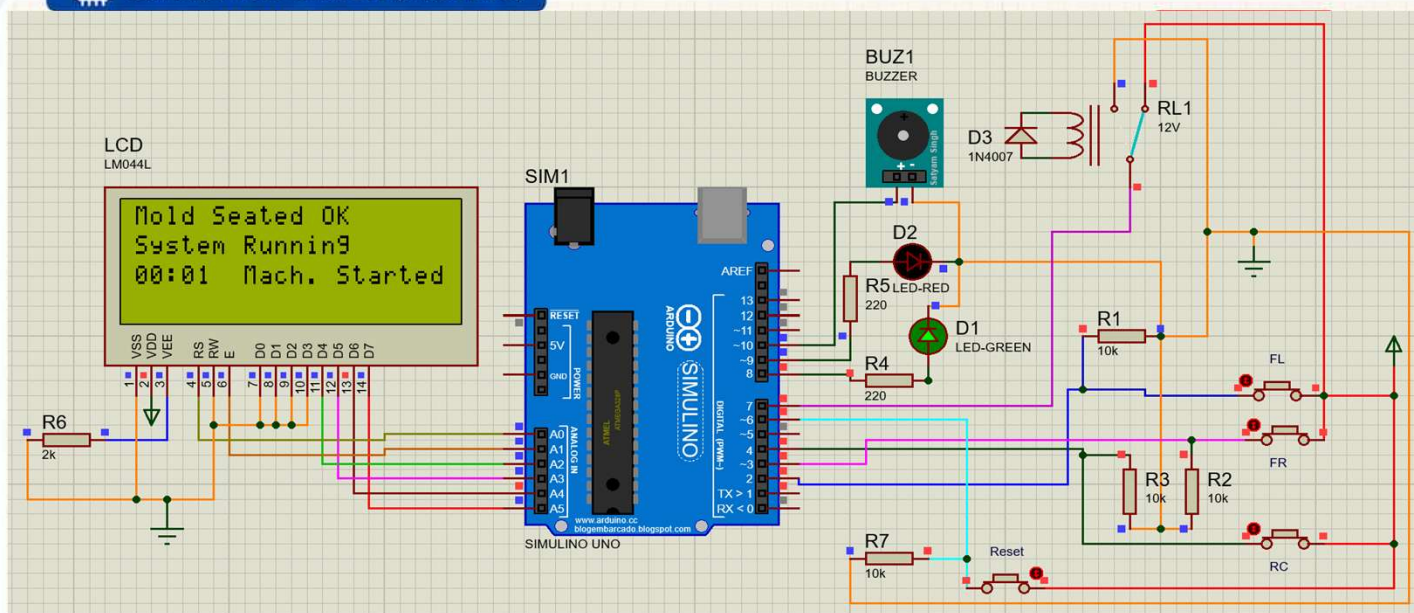
システムこうぞう

System Design & Architecture

Hardware architecture and circuit design of the mold verification and safety interlock system



SIMULATION CIRCUIT (PROTEUS)



KEY COMPONENTS



Controller (Arduino UNO)
Processes inputs and controls all outputs.



Display (LCD 16x2)
Shows system status and messages.



Status Indicators
Green (Ready) and Red (Fault) LEDs indicate system state.



Buzzer
Provides audible alert for fault conditions.



Relay (RL1)
12V relay enables the machine only when mold is verified.



Sensors / Switches
FL, FR, RC verify mold position at three critical points.



Reset Switch
Manual reset to acknowledge and clear faults.



Safety Logic
Machine will start only when all three sensors are activated. Any fault blocks the machine and triggers an alert.

THREE-POINT VERIFICATION

FL



Front-Left (FL)

Verifies mold presence at the front-left position.

FR



Front-Right (FR)

Verifies mold presence at the front-right position.

RC



Rear-Center (RC)

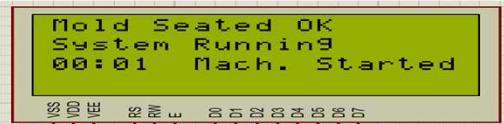
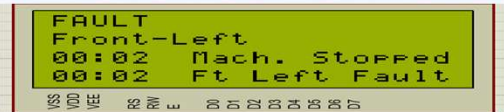
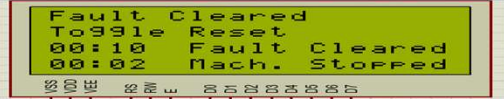
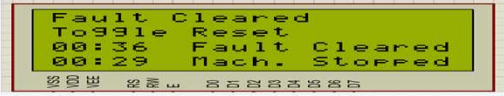
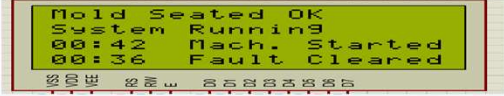
Verifies mold presence at the rear-center position.



シミュレーション テスト結果 SYSTEM TEST RESULTS

Timeline reconstructed from actual simulation screenshots



STEP	TIME	LCD DISPLAY (LM044L)	EVENT	VERIFIED BEHAVIOUR
1	00.01		 Machine started	 Normal start
2	00.02		 Front-Left fault	 Correct sensor named, stopped
3	00.10		 Fault cleared	 Stayed locked, did not resume
4	00.21		 Machine started	 Deliberate restart, 18s gap
5	00.29		 Front-Right fault	 Correct sensor named, stopped
6	00.36		 Fault cleared	 Stayed locked, did not resume
7	00.42		 Machine started	 Deliberate restart, 10s gap



All test cases validated successfully in Proteus simulation. System correctly detects faults, locks the machine, and requires deliberate restart after issue clearance.



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Conclusion & Future Scope

A **Safer** Process Today, A **Smarter** Factory Tomorrow

PROJECT OUTCOME



Safe Startup

Machine starts only when all conditions are satisfied.



Mold Verified

Three-point verification ensures correct mold placement every time.



Machine Protected

Fault detection and interlocks prevent accidents and damage.


FUTURE SCOPE

Building Safer, Smarter Manufacturing

Driving Productivity • Ensuring Safety • Enabling Intelligence



ありがとうございます

Thank You!



Mold Verification and Safety Interlock System



Safe
Operation



Reliable
Performance



Improved
Productivity



Future
Ready

“ Ensuring Safety. Enhancing Reliability. ”
Building a Smarter Tomorrow.

